

Project Report

on

AI BASED FACIAL RECOGNITION SYSTEM FOR SECURE ENTRY IN DEFENCE ZONE

Submitted in partial fulfillment of requirements for the award of the degree

Bachelor of Technology

In

Artificial Intelligence and Machine Learning

To

IKG Punjab Technical University, Jalandhar

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Semester: 6th

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Department of Computer Science & Engineering

CGC – College of Engineering, Landran

Mohali, Punjab – 140307

April 2026

CANDIDATE'S DECLARATION

I hereby declare that the work which is being presented in the 6th semester, entitled “**AI based facial recognition system for secure entry in Defence zone**”, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence and Machine Learning and submitted to CGC College of Engineering, Landran, Mohali is an original piece of work carried out by me during the period from January 2026 to May 2026 under the supervision of Ms. Sukhdeep Kaur, department of CSE, CGC College of Engineering Landran Mohali Punjab India.

The matter embodied in this File has not been submitted by me for the award of any other degree of any other University/Institute.

Gurnoor Singh

The File entitled “**AI based facial recognition system for secure entry in Defence Zone**” submitted by **Gurnoor Singh** to Department of CSE for the award of the degree of Bachelor of Technology is a Bonafide record of the project work carried out by him/her under my supervision and guidance. This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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Certificate

I hereby declare that the 6th Semester Training Report entitled “**AI based facial recognition system for secure entry in Defence Zone**” is an authentic record of my own work as requirements of 6th sem academic during the period from January 2026 to May 2026 for the award of degree of B. Tech. in AIML, CGC-College of Engineering, Landran, Mohali under the guidance of Ms. Sukhdeep Kaur.

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2320401

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Protocol
CSS	Cascading Style Sheets
CV	Computer Vision
DL	Deep Learning
FR	Face Recognition
GUI	Graphical User interface
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IoT	Internet of Things
JS	JavaScript
ML	Machine Learning
SQL	Structured Query Language
UI	User Interface
YOLO	You Only Look Once

ABSTRACT

Defence zones and restricted areas require very high security. Traditional methods such as ID cards, passwords or manual verification are used for entry to these zones. However, these methods have several problems. ID cards can be stolen, duplicated or misplaced. Passwords can be shared. Manual checking can make human errors and is also time-consuming. Therefore, there is a need for more reliable and automated entry/access method.

The system will use Artificial Intelligence and computer vision techniques to detect and recognize faces. The proposed system will use a camera to capture and analyze the face of the person at the entry point. The face of the individual will work as the ID card or token. Then the system will analyze the face and convert the captured face into numerical data called face encodings. The face encodings of authorized personnel will be stored in a database. When the camera captures a face, the face will be converted into face encoding and then compared to stored data in the database. Access will only be granted if face matches the stored data otherwise entry will be denied and security will be alerted.

The system uses Python, OpenCV and face_recognition library to detect and recognize faces, and uses Flask to develop web-based interface. The system operates in real time providing quick and accurate identification of people. It also maintains logs of entries and exits, which helps in monitoring and security analysis. It will help to improve security and prevent unauthorized entries. This will make the whole area much safer as it will not rely on humans which can make mistakes. This type of system can be used in areas like defence zones, airports, government buildings and other high security areas.

The proposed system was successfully implemented and tested in real time conditions. The system was able to correctly identify individuals and deny the access of unauthorized users with quick response time. The results demonstrate that the system can improve security, reduce human effort and provide real time solution for secure entry management in defense and other high security areas.

Chapter – 1

Introduction

In areas such as defence zones, military bases, and other restricted locations security is a major concern. Entry of only authorized personnel should be allowed in these areas. Traditional entry methods use ID cards, passwords or manual verification by security personnel. But these methods have several limitations and risks. ID cards can be stolen, lost, or duplicated. Passwords can be shared and Manual verification may lead to errors and delays.

Therefore, there was a need for more reliable and automated system. Modern Technology is being used to improve the security systems. Methods such as biometric authentication is a reliable method of identifying individuals. Facial recognition is widely used for identification processes. Facial recognition systems are used widely in security, surveillance and access control applications.

1.1 Facial Recognition System

Facial recognition is a biometric identification technology which identifies a person using their facial features. It identifies individuals using features such as eyes, nose shape, jawline and lip shape.

The system captures the image of a person's face using a camera. The system converts facial features into digital data called face encodings. These encodings are unique to each individual. Facial features of the authorized personnels are extracted and stored in a database for identification.

If the person is identified, access is granted otherwise denied and security is informed. This technology is faster, more secure and more reliable than traditional form of system for entry identification.

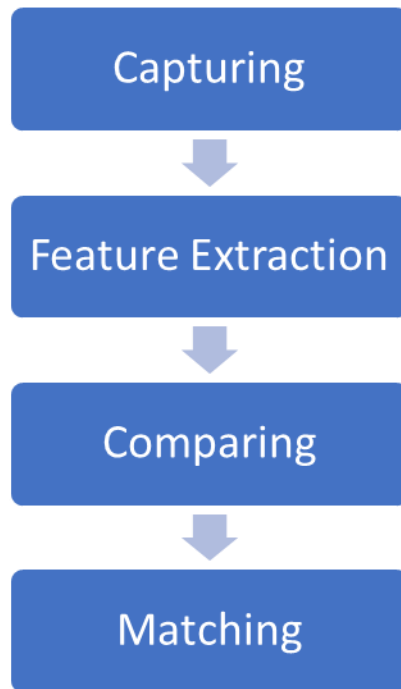


Figure 1.1: Basic Facial Recognition Process

1.2 Types of Face Recognition Technologies

Face recognition is implemented using various approaches depending on the techniques used for detecting and identifying facial features.

1.2.1 Feature Based Methods:

It identifies faces using distinct facial features such as eyes, nose, and mouth. These methods focus on measuring the spatial relationships between these features, such as the distance between the eyes, the width of the nose, and the shape of the facial structure. It is one of the earliest face recognition techniques.

1.2.2 Appearance Based Method:

This method identifies faces by analyzing the overall appearance of the face image rather than focusing on individual features. This method converts the entire face into mathematical representation using statistical techniques.

Techniques such as Eigenfaces and Fisherfaces are commonly used, where facial images are transformed into vectors and important features are extracted based on variations in pixel

intensity. The system compares these representations with stored data to recognize individuals.

1.2.3 Deep Learning Based Models:

It uses neural networks to recognize faces. It learns facial patterns from large image datasets. These models are trained on large datasets of facial images, allowing them to learn complex patterns and features such as facial structure, texture, and expressions automatically. It provides high accuracy in modern systems.

1.2.4 Hybrid Models:

It combines feature-based and appearance-based approaches to improve recognition accuracy and reliability. These methods use geometric features such as the distance between facial landmarks along with the overall facial appearance to create a more comprehensive representation of the face.

1.3 Applications of Facial Recognition System

Facial recognition technology is used in various fields for identification, authentication and security. It is a fast method to check someone's identity.

One of the biggest applications is in defence and security systems. Facial recognition helps control who gets into areas and prevent unauthorized entries. Airports and border control also use it to check passengers and keep an eye on people.

In law enforcement, facial recognition helps in identifying suspects and tracking criminal activities. It is also used in banking and financial institutions for secure authentication and fraud prevention.

This technology is also used in smartphones and personal devices for unlocking devices and user authentication.

It is also used in surveillance systems, attendance systems and smart cities projects to improve monitoring and management. Due to its efficiency and reliability, it has been an important tool in modern security and identification systems.

1.4 Advantages of Facial Recognition System

Facial recognition systems have several advantages over traditional recognition systems. One of the advantages is contactless authentication system where there is no need of any physical

interaction, unlike ID cards or manual verification.

Another advantage is speed and efficiency as system can identify individuals in real time. It also offers high accuracy.

Facial recognition systems are difficult to forge or duplicate, making them more secure than traditional methods such as passwords or ID cards. Additionally, these systems can be easily integrated with existing surveillance infrastructure, making them a scalable and cost-effective solution.

1.5 Limitations of Traditional Security Systems

Traditional security systems such as ID cards, passwords and manual verification have several limitations. One of the major disadvantages of ID cards is that it can be lost, stolen or duplicated which can also lead to unauthorized access in defence zones.

Similarly, passwords can be shared. Manual verification is very time consuming and can also lead to human errors. It leads to delays and can become inefficient in high-traffic areas. Manual verification requires human manpower and also supervision, which leads to additional costs.

These limitations highlight the need for more advanced and automated security solutions. Biometric identification techniques such as face recognition address these issues by providing more secure and efficient method of identification.

Summary

In this chapter the concept of facial recognition system and its importance is discussed. Various types of facial recognition methods, including feature-based, appearance-based, deep learning-based, and hybrid approaches, have been explained. The applications and advantages of facial recognition along with disadvantages of traditional security systems have been highlighted. This forms the basis for the proposed system for secure entry in defence zones.

Chapter-2

Literature Survey

Literature survey is an important part of any research or project as it helps in understanding the existing studies and technologies related to the proposed system. It provides overview of already been carried out researches by different researchers in the field of facial recognition, biometric identification, computer vision and artificial intelligence. By studying the previous papers, it helps to understand the approaches that has been taken before to solve similar problems.

In recent years, artificial intelligence and computer vision has been widely used for security systems, surveillance, and access control mechanisms. Many researchers have suggested methods for detecting faces recognizing faces and monitoring people automatically in defense and high-security areas. These systems will make the security better and reduce the human effort. It will help to identify people more accurately. Automated systems will help to make the process fast.

The literature survey helps to understand the strengths and limitations of the existing systems. It also highlights the challenges faced by them and ways of possible improvement. In the following section, different research papers related to facial recognition system, artificial intelligence and security applications are reviewed.

2.1 Existing Research Works

In recent years, the push to secure defense zones has led to a shift from manual guarding to using AI-driven facial recognition and surveillance. Many researchers have looked at how to make these systems faster and more reliable in tough conditions.

2.1.1 Advanced Detection and Real-Time Processing

One of the biggest hurdles in defense is making sure AI can see clearly in the dark or through bad weather. Chutke et al. [1] tackled this by creating a system that combines regular video with thermal and night-vision feeds. By using powerful models like YOLOv8, their system can spot weapons or unauthorized people even when visibility is poor. They also used something called a Kalman Filter to help the system track moving targets more smoothly and

reduce the number of annoying "false alarms."

2.1.2 Smart Border and Parameter Security

Securing a long border or a large military perimeter is a massive job. Ige et al. [2] looked into using OpenCV to build a model that doesn't just watch the gate, but actually detects people from far away. Their approach is interesting because it automatically snaps a photo of the person's face and checks it against global databases before they even reach the entry point. This acts as an early warning system to stop potential threats before they get close. Building on this, Saadi and Samir [3] analyzed how AI has become a global standard for border monitoring. However, they pointed out that environmental factors—like the specific geography found in regions like Algeria—can interfere with how well these cameras work, suggesting that AI models must be adapted to the specific terrain they are guarding.

2.1.3 Improving Entry Point Efficiency

Military bases often face huge traffic jams due to slow, manual ID checks. Walker [4] studied this problem and found that manual security is not only slow but also takes up too many personnel. By simulating an automated facial recognition entry point, Walker proved that it is possible to significantly speed up the flow of people and vehicles without making the base any less secure. This research supports the idea that AI is necessary for both security and operational efficiency.

2.1.4 Algorithmic Frameworks and Ethical Challenges

As the technology gets better, the math behind it becomes more complex. Chalakov et al. [5] examined how Deep Neural Networks (DNN) and Support Vector Machines (SVM) are used to identify personnel in defense HR systems. While these algorithms are highly accurate, the authors warn that there are significant ethical and legal challenges regarding data privacy. For a facial recognition system to be successful in a modern defense zone, it must balance high speed recognition with strict data protection laws.

2.1.5 Balancing Security and Ethics

Javed et al. [6] highlight that while AI significantly boosts the efficiency and predictive ability of border and entry control, it must be balanced with transparency. This research justifies the "Audit Trail" and "Logging" features of our proposed system, ensuring that while

the system is automated, it remains accountable and ethical in a defense context.

Table 2.1: Comparative Study of existing Facial Recognition Systems

Author / Paper	Technique Used	Advantages	Limitations
Chutke et al. [1]	Basic Face Recognition	Simple implementation	Low accuracy in real-time
Sharma et al. [2]	AI-Based Recognition	High accuracy	High computational cost
Kumar et al. [3]	Machine Learning Approach	Efficient processing	Affected by lighting conditions
Singh et al. [4]	Deep Learning Model	Better performance	Requires large dataset
Proposed System	OpenCV + face_recognition	Real-time, efficient, easy to use	Slightly affected by lighting and occlusion

2.2 Research Gaps

Despite the progress there exist several critical gaps still exist in high-security defense zones:

- **Vulnerability to Advanced Spoofing:**

Most current systems can stop simple photo-based attacks but they still struggle with sophisticated deepfake masks or synchronized digital-physical attacks that fool both liveness detection and facial recognition systems.

Facial recognition systems are still vulnerable to these spoofing attacks.

- **Latency in Edge Deployment:**

High-accuracy models are often too heavy to run on local gate cameras. This creates a gap between achieving accuracy and maintaining real-time speed for fast-moving military checkpoints. We need facial recognition systems that work in time.

- **Performance in Unconstrained Conditions:**

Systems handling variables at once such as a soldier wearing a tactical mask in heavy rain or moving at high speed lack robustness.

Facial recognition systems need to work in various conditions.

- **The Black Box Problem:**

Current AI often provides a Match or No Match result without explanation.

In defense zones security personnel lack AI to understand why an authorized individual was rejected.

Summary

The research shows that while facial recognition is common, defense zones need a unique mix of speed and high security. OpenCV and ML are one of the best ways for face recognition in most environment. Different environment can affect the working of the system. Switching from manual checks to AI-driven encodings that will make the security much tighter and less dependent on human guards. The system tries to make the overall process automated rather than handled by humans.

Chapter – 3

Problem Statement and Objectives

3.1 Problem Statement

Defence zones and restricted areas require very high level of security to prevent unauthorized access. So, the entry must be strictly controlled. However, traditional access techniques such as ID cards, passwords and manual verification are still used in some high-security places. These methods are inefficient and time-consuming.

ID cards can be stolen, shared and duplicated. Passwords can be shared and manual verification requires manpower and still can lead to delays and errors. In defence zones, even a small breach can result in serious consequences, making these traditional methods unreliable.

Therefore, there is a need for an automated and reliable identification system which identifies individuals in real-time.

Facial recognition technology helps to overcome the challenges of traditional security systems. It identifies the individuals using biometric features. It is contactless and efficient, maintain the logs of entries and exits.

The goal of this project is to create a facial recognition system that uses AI to identify and let people enter defence zones without human intervention improving the overall security.

Table 3.1: Issues in Traditional Security System

Issues	Description
ID Card Theft	ID cards can be stolen, duplicated or lost
Password Sharing	Passwords can be shared among users
Human Error	Manual verification may lead to mistakes
Time Consumption	Entry process is slow and inefficient
Security Risk	Unauthorized access is possible

3.2 Objectives

The main objectives of the Proposed System are:

- To study the fundamentals of facial recognition technology and understand how ML and CV techniques are used in identifying faces.
- To design a secure and efficient system architecture capable of identifying and recognizing faces.
- To develop a working face recognition system that performs operations in real time.

3.3 Scope of the Project

The scope of the project is to develop an AI-based facial recognition system for secure, efficient and automated entries in restricted zones. The system will use a camera to capture facial images and process them to generate face encodings. These encodings are then compared to stored encodings of authorized personnel in database to verify identity.

On the basis of comparison, the system grants access to authorized personnel and denies entry to unauthorized personnel. The denied attempt will notify the security team to take action.

This project focuses on implementing a real-time facial recognition system using ML and computer vision techniques. It aims to reduce manual effort and improve the efficiency and accuracy of access control mechanisms.

This system can be used in high-security environments such as defence zones, government buildings and other restricted areas.

3.4 Need for the Project

Security management is very important in defence zones, military bases and other restricted areas. They require strict control over who enters and exits these zones. Any security breach in such areas could have high consequences, including threats to national security, loss of confidentiality information and potential risk to human lives. Therefore, it is essential to have high security in these zones.

Traditional access techniques such as ID cards, passwords and manual verification are still

used in some high-security places. These methods are inefficient and time-consuming and have several limitations. ID cards can be easily lost, stolen, or duplicated, making them unreliable for high-security applications. Password-based systems are also vulnerable, as passwords can be shared, guessed, or hacked. Manual verification by security personnel is time-consuming and prone to human error. It can lead to delays, especially in areas with heavy traffic, and may not always guarantee accurate identification.

In modern secure environments, there is a need for automated systems that can provide faster, accurate and reliable identification of people. These systems remove human dependency and reduce chances of errors. They also provide real-time decision-making, which is crucial in high-security areas.

The increasing advancement in technology has made it possible to develop a security system that can operate in real time with minimal human intervention. Such security systems provide high accuracy and ensure faster processing. They provide better monitoring capabilities, making them highly suitable for sensitive environments like defence zones.

Biometric technologies provide an effective solution for identification and verification purposes. This technology uses features such as fingerprints, iris patterns, facial features for authentication. This type of technology leads to a contactless system where no physical interaction is needed. It can also work with the existing camera infrastructure of the defence zones.

Facial recognition systems use AI and computer vision to detect and identify faces. The faces are converted to facial encodings which can be compared to identified faces in real-time and provide/deny access to the individual.

This approach not only improve security but also enhances the efficiency of the process. There is no need of manual checking and ensure only authorized personnels can access those areas. Furthermore, it provides a scalable solution that can be implemented in various high-security environments.

In conclusion, there is a strong need for an automated and intelligent access control system that can overcome the limitations of traditional methods. The integration of facial recognition technology in security systems offers a reliable, efficient, and future-ready solution for ensuring secure entry and monitoring in restricted areas.

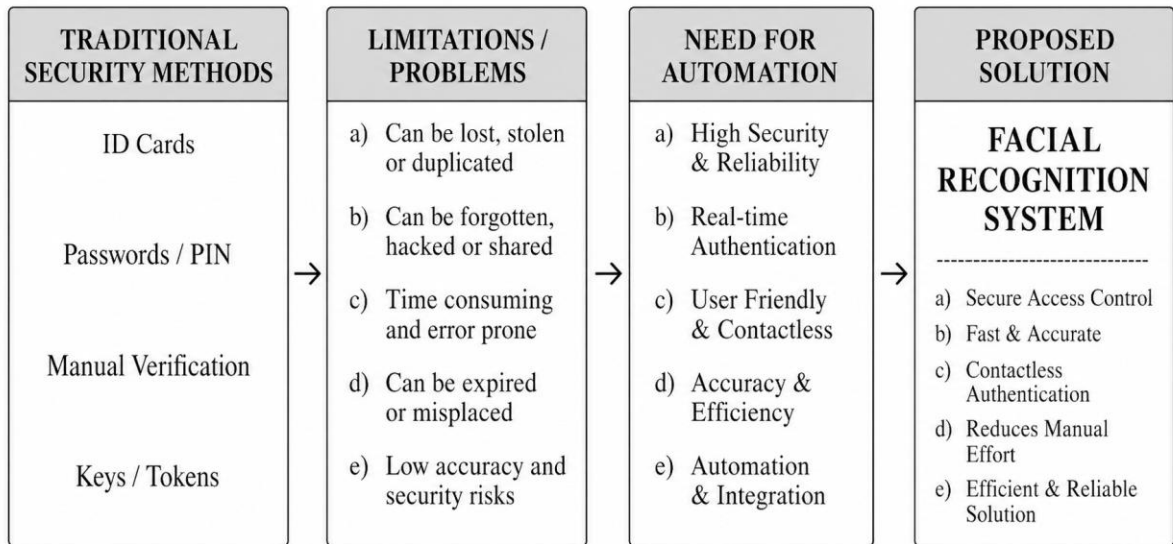


Figure 3.1: Need for Automated Facial Recognition System

Summary

In this chapter, the limitations of traditional security systems have been discussed. These methods are prone to issues like theft, duplication, human errors, and time delays, making them less reliable for high-security environments such as defence zones. So, the need for a more secure, efficient and automated system has been highlighted. The proposed system uses ML and CV to overcome the limitations and provide accurate and real time results. This approach reduces human dependency and enhances overall security and efficiency in restricted areas.

Chapter – 4

Methodology and Implementation

4.1 System Overview

The proposed system is an AI based face recognition system designed to improve security at the defence and restricted areas. This system will make the process automated and efficient. This system integrates computer vision and ML techniques with a web-based interface to enable real time identification.

This system will use a camera that captures the faces of individuals at the entry point. These images are process to detect and recognize faces using the OpenCV and face_recognition libraries. The captured facial details are converted into digital data which is called face encodings which will be compared to stored face encodings. The face encodings of authorized people are stored in the database.

The proposed system is implemented using Python and uses Flask framework to provide a web-based interface. The interface allows the user to open the camera, scan faces and view recognition results in real-time.

The integration of technologies provides seamless communication between different components of the system. It provides efficient data processing from capturing to recognition, enabling the system to operate smoothly in real time.

The overall workflow of the system includes image acquisition, face detection, feature extraction, face recognition and access control. Based on the comparison, the system grants access to authorized individuals and denies entry to unauthorized individuals. An unauthorized attempt will alert the security.

The system will maintain logs of all entry and exit activities which can be used for monitoring and analysis. The integration of front end, back end and database management makes this system a complete and efficient solution for secure and efficient access in high security environments.

Table 4.1: Modules and Their Functions

Module	Function
Image Acquisition	Captures images from the camera
Face Detection	Detects faces in captured frames
Feature Extraction	Extracts facial features & converts into encodings
Face Recognition	Matches captured face with stored data
Database Management	Stores user data and facial encodings
Access Control	Grants or denies access based on recognition

4.2 Image Acquisition

This is the first step in the face recognition system. The system will capture real-time images of individuals using a webcam or a camera installed at entry point. The camera will continuously capture a video which is processed as a sequence of frames for further analysis. Flask is used to provide a web-based interface. Through the interface, the user can activate the camera. The captured video will be divided into frames, which will then be used for face detection and recognition.

Proper image acquisition is very important for accurate face recognition. Factors such as light conditions, camera resolution and face orientation can affect the quality of captured images. So, camera must be position in such a way that it captures a front-facing images of individuals at the entry point.

The captured images will be forwarded to the next stage in the system where face detection will be performed

4.3 Face Detection

Face Detection is an important step in face recognition process. This process identifies and isolates human faces from the captured images. The frames from previous steps will be used to identify the face of the individual.

The proposed system will use OpenCV and face_recognition libraries for detecting faces in real-time frames. These libraries use ML to find a face in the frame. They do this by looking for a pattern such as shape of eyes, shape of mouth, etc. Once the face is detected, the system marks the region as region of interest using bounding boxes.

Instead of focusing on the entire image, the system only focuses on the of the individual. This improves efficiency and speed of the system. The detected face is extracted and passed to the next process.

The system is capable of detecting multiple faces in a single frame and process each face individually. It also works under different lighting conditions and angles.

Accurate face detection is essential for overall performance of the system as any error can cause incorrect identification.

4.4 Facial Feature Extraction

Facial feature extraction is the process of converting the detected face into digital data that can be used for identification. After the face is detected, the system will analyse the face to extract unique features such as shape of eyes, nose, jawline, etc.

face_recognition library is used for this process and it will generate face encodings for each detected face. These encodings are high dimensional numerical vectors that uniquely represent the facial feature of an individual. Unlike raw images, these are compact and efficient to store and use.

A trained model is used to extract key facial features and convert them into encodings. These encodings act as digital representation of a person's face. These encodings are also used to compare them with the data in database.

Feature extraction plays an important role in ensuring accurate identification. Face encodings are better than raw image stay stable under variations such as lighting, minor facial expressions and slight changes in angle.

The extracted encodings are used in next stage where they are compared to stored encodings in database.

4.5 Face Recognition and Matching

Face recognition and matching is the step in which the system identifies an individual by comparing their extracted face encodings with the stored face encodings in the database. This determines if the person is authorized or a not.

In the proposed system, the face_recognition library is used for comparison between stored and captured face encodings. Similarity is calculated between the two face encodings using distance-based methods.

A threshold level is defined, If the similarity is below that then the person is unauthorized otherwise access is granted. The recognition process is performed in real-time and allow quick identification of individuals.

Accurate matching is essential for ensuring reliability. By using a predefined threshold, the system minimizes false positive and false negatives improving overall performance.

The result of this stage decides whether to grant or deny entry on the recognition outcome.

4.6 Database Management

Database plays an important role in storing the facial encodings of authorized individuals. Database management helps in retrieving that information.

In the proposed system, Facial encodings are stored in the database using the pickle module. The file contains encoded representation of faces of authorized personnels. These encodings help in fast loading and efficient comparison during the recognition process.

The details of authorized personnels including name, role, department and clearance levels are also stored in the database. Database helps to retrieve relevant data once a face is successfully recognized.

During operation, when a phase is captured and encoded, the system compares the encoded phase with the stored encodings from the pickle file. If the face matches with some data on database, The corresponding user information is fetched from database and displays on interface.

The system also maintains a log of entries and exits activities. Efficient database management has in quick retrieval of data, accurate recognition and smooth performance.

4.7 Access Control Mechanism

The access control mechanism is responsible for making the final decision regarding whether the individual is authorized or not. On the basis of this decision his will be allowed or denied in the restricted area. The decision is on the results obtained from the face recognition and matching stage.

Once a face is recognized, the system will retrieve the details from the database. Only individual whose facial data matches the stored encodings and who made the required authorization criteria are granted access.

Is this captured face data matches with some phase in database, the system displays “Access

Granted” confirming the access. If no match is found, the system will consider that individual as unauthorized and display an “Access Denied” message. In this case the security will be informed.

This process happens in real time ensuring instant decision making whenever the camera captures a face. This allows smooth and efficient entry process in the defence zones.

A log will be maintained in the database which stores each successful or unsuccessful attempt. This allows monitoring and helps in identifying suspicious activities.

4.8 System Workflow

The system Workflow represents the sequence steps performed by facial recognition system, from capturing the image to granting or denying access.

The process begins with the camera capturing live video and extracting frames from it. These frames are used to detect presence of faces using computer vision technologies.

Once a phase is detected, the system will extract facial features from that and convert it into digital representation known as face encodings. This encoding is then compared with stored data in the database. If a match is found, the system will grant the access to that individual otherwise the entry will be denied and the security will be alerted.

All access, whether successful or unsuccessful are recorded in system logs along with details such as timestamp and access status. The system Continuously monitors the video stream for new faces enabling real time operation.

This workflow ensures that the system performs real-time, efficient and accurate identification.

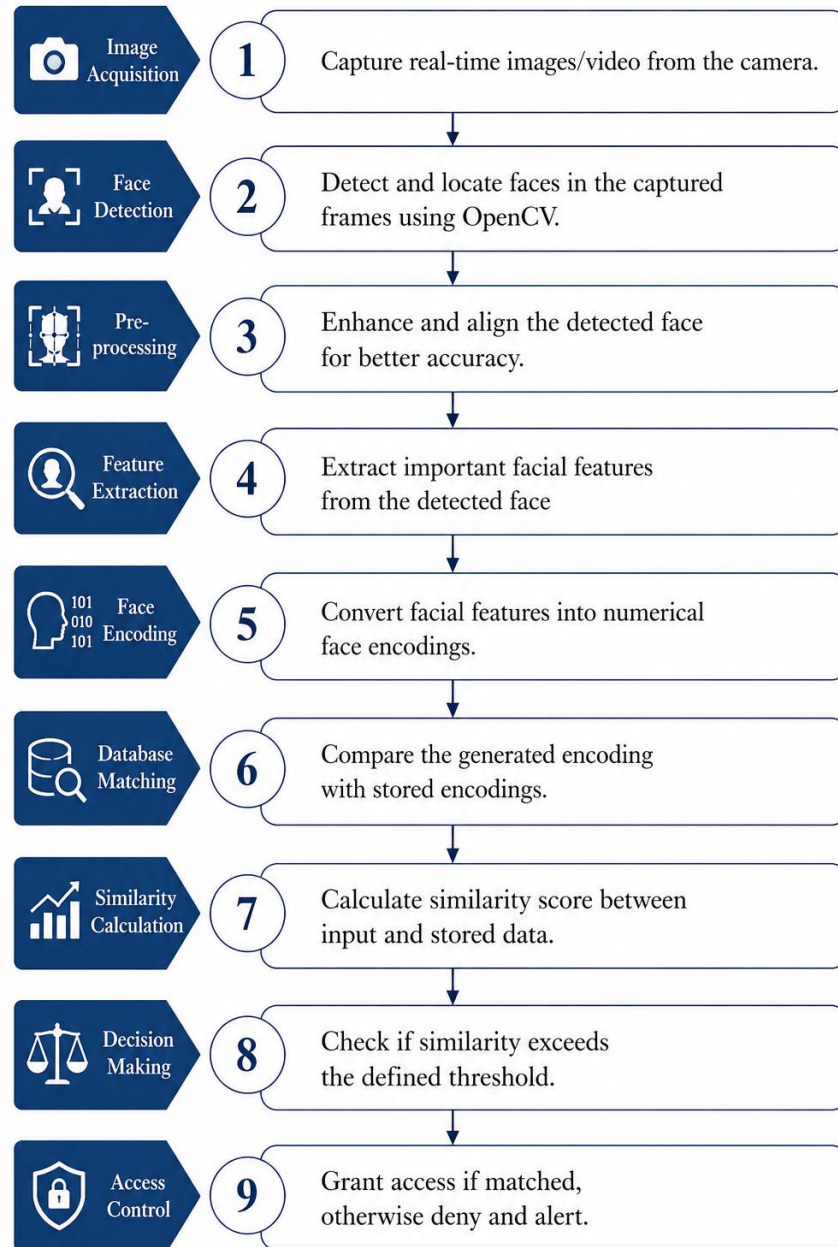


Figure 4.1: Workflow of the Proposed Facial Recognition System

4.9 Web Interface and User Interaction

The proposed system has a web-based interface that allows user to interact with the facial recognition system. The interface is developed using Flask framework for back end, while using HTML, CSS and JavaScript for developing front end.

The interface provides us with controls to start and manage camera for capturing facial

images. The user can initiate the face recognition process through a scan button which captures frame from the live video stream and sends it to back end for processing. Automatic scanning is also available.

Once the system processes the captured image, the results are displayed on the interface. The system will also show relevant details such as name role and access status of the individual.

The interface also displays log of entry and exit activities. These logs contain information such as identity of individual, timestamp and access status.

The integration of front end and back end ensures more communication between user interface and processing system. This makes the overall system interactive, user friendly and suitable for real time deployment in high security environment.

4.10 Technologies Used

The development of the proposed facial recognition system involves integration of various technologies and tools for efficient implementation. These technologies are used for image processing, facial recognition, data management and user interaction interface development.

4.10.1 Python

It is the primary programming language used for developing the proposed system. It is used to provide a simple, readable and efficient environment to implement ML and computer vision. It supports a wide range of libraries and frameworks, which makes the implementation easy.

Python is used to integrate different components of the project together. Python libraries such as OpenCV and face_recognition are used to perform face detection and recognition tasks. Flask framework is used to create a web-based interface for user interaction. It provides connection between front end and back end of the system.

Due to its versatility, wide range of libraries, and ease of implementation, Python is widely used for development of intelligent systems

4.10.2 OpenCV

OpenCV (Open Source Computer Vision Library) is an open-source library used for image processing and computer vision tasks. In the proposed system, OpenCV is used to capture video frames from the camera and perform real time face detection.

This library the system to find faces in the frames. It uses pre trained models and algorithms

to detect faces based on patterns such as arrangement of eyes, nose and mouth.

Open CV as in faces and drawing bounding boxes around detected faces. It helps to isolate the facial areas for further processing.

It is used to increase the efficiency and real time processing capabilities. Thus, it is used in various computer vision applications.

4.10.3 face_recognition Library

The face_recognition library is a Python library used for performing facial recognition and identification task. It is built on top of the dlib machine learning toolkit and provide simple and efficient method for detecting faces from extracting features and recognizing individuals. In the proposed system, this library is used to generate face encodings which are numerical digital representations of facial features. These encoding our unique identification or each individual and are used for comparison with store data in database.

The face recognition library is known for its high accuracy and ease of use, making it suitable for real time applications such as face recognition system.

4.10.4 Flask Framework

It is a lightweight web framework used for developing the back end of applications. It provides a simple and flexible environment for building web-based applications using Python.

In this project, Flask is used to create a web-based interface which will allow the user to interact with the proposed system. It will handle request from frontend, process the data and return the results to the user interface. It acts as a bridge between frontend and backend processes. It also enables real-time communication between user interface and the system.

Due to its simplicity, efficiency and ease of integration with Python libraries, Flask has been used for the proposed system.

4.10.5 Frontend Technologies (HTML, CSS, JavaScript)

The frontend of this proposed system is developed using HTML, CSS and JavaScript which integrated together provide an interactive and user-friendly interface. These technologies are used design, style and handle user interactions.

HTML (HyperText Markup Language) is used to create the structure of web pages. It can include elements such as buttons, video display area and result panels. CSS (Cascading Style

Sheets) is used to enhance the visual appearance of the interface created using HTML. JavaScript is used to make the system interactive. It will allow functionality such as starting the camera, capturing frames and sending it to the back end for processing. It has system to become dynamic.

4.10.6 SQLite Database

SQLite is a lightweight, file-based relational database using the proposed system to store and manage data of authorized personnel and access logs. They do not require any separate server, making it simple and efficient for small to medium scale applications.

In this project, SQLite is used to store details of the authorized individuals. During the recognition process, once the phase is matched the system will retrieve the corresponding user interface from the database and display it on the user interface.

The database will also maintain logs of entry and exit activities. SQLite provides fast data retrieval and easy integration with Python making it suitable for this face recognition system.

4.10.7 Pickle Module

The pickle module is a Python library used for serializing and storing data in binary file format. In the proposed system, it is used to facial encodings generated by face_recognition library.

The face encodings of authorized personnel are saved in encodings.pkl file using pickle module. It allows to efficiently load and access the encoded data during recognition process.

When a new face is captured by the system, it's encoding is generated and compared with stored encoding loaded from the pickle file. This helps to increase the speed and efficiency of the system by reducing the processing time.

Pickle module is used in this project as it provides a simple and efficient way to handle large sets of facial data, it an important component of the overall system.

4.10.8 NumPy

NumPy (Numerical Python) is a fundamental library used for numerical computations in Python. It is used to handle errors and perform mathematical operations on the image data. The face encodings are stored in numerical format. NumPy enables efficient computation and supports operations required for comparing facial features. It provides high performance and ability to handle large data sets making it suitable for real time applications.

4.10.9 Pillow

Below is a Python library used for image processing and manipulation. In the purposed system, it is used handle image format and perform basic image operations such as cropping the image. It helps in preparing the images for further processing.

It ensures that the image is in correct format and quality before being analysed.

Table 4.2: Technologies Used in the System

Technology	Description	Purpose
Python	High-level Programming Language	Used for overall system development
OpenCV	Computer vision library	Used for face detection and image processing
face_recognition	Python library based on dlib	Used for face encoding and matching
Flask	Web framework	Used for backend development and server handling
HTML	Markup language	Used for structuring web interface
CSS	Styling language	Used for designing and layout of interface
JavaScript	Scripting language	Used for frontend interactivity
SQLite	Lightweight database	Used for storing user data and logs
Pickle	Python module	Used for storing facial encodings
NumPy	Numerical library	Used for array and mathematical operations
Pillow	Image processing library	Used for handling image formats

4.11 System Requirements

The implementation of the proposed facial recognition system requires specific hardware and software components to ensure its proper functioning. The requirements are necessary for capturing images, processing data and running the system efficiently in real time.

4.11.1 Hardware Requirements

For the proposed system, the hardware requirements are minimal and can be fulfilled using standard computing devices. A system basic configuration computer is sufficient to run the application efficiently.

The following are the hardware components are required:

- A computer or laptop with a minimum of 4 GB RAM
- A processor such as Intel i3 or higher
- A webcam or built in-camera for capturing facial images
- Storage space for storing datasets and logs

These components ensure smooth execution of the system and enable real-time process.

4.11.2 Software Requirements

The proposed system requires various software tools and libraries for its development and execution. These software components are essential for its proper working.

The following software requirements are needed:

- Operating system: Windows
- Programming Language: Python (version 3.9 or above)
- Libraries: OpenCV, face_recognition, NumPy, Pillow
- Framework: Flask
- Database: SQLite
- Web Technologies: HTML, CSS, JavaScript

These software components work together to support the development and execution of the facial recognition system.

4.12 System Architecture

The system architecture represents the overall structure of the proposed facial recognition

system and shows how different components interact with each other. It provides level by level view of flow of data from input to output.

The proposed facial recognition system follows a client-server architecture. The frontend act as a client and the backend act as a server. HTML, CSS and JavaScript are used to develop the frontend. It provides a user interface through which the user interacts with the system. Flask is used to develop the backend of this project. It is used to handle data processing and communication with various modules

The process begins with the camera capturing live video, which is then sent to backend in form of frames. OpenCV processes the frames for face detection. Once a face is detected, the system uses face recognition library to extract facial features and generate face encodings.

The face encodings of authorized individuals are stored in a database. They generated encodings are compared with the stored encodings loaded from the pickle file. If a match is found, the system retrieves the corresponding user details from the database. On the basis of this process, the access control module decides whether to grant or deny entry.

The system maintains a log of access attempt in the database. Each successful and unsuccessful attempt is recorded.

The integration of frontend, backend, database and ML modules ensure smooth and efficient operation of the system.

4.13 Module Description

The proposed facial recognition system is divided into several modules, each having a specific task. These modules work together to ensure accurate identification and secure access control. The modular design improves organization, maintainability and efficiency.

4.13.1 Image Acquisition Module

The image acquisition module is responsible for capturing facial images of individuals using a camera. The proposed system uses a webcam to capture live video which is processed as a sequence of frames.

This module is integrated with web-based interface developed using Flask, allowing the user to control camera through the browser. The user can use the camera to initiate the scanning process using the interface. Each captured frame is processed in real time and passed to face

detection module. The quality of the image plays an important role in overall accuracy of the system. Poor lighting conditions or incorrect positioning of the camera will reduce the accuracy.

4.13.2 Face Detection Module

This module is responsible for identifying the presence and location of human faces in the captured frames. This module reviews each to determine whether a face is present.

Open CV and face recognition libraries are used to detect faces. These libraries are pre-trained ML models to detect facial regions by analysing patterns such as arrangement of eyes, nose and mouth. Once a face is detected, the system highlights it using bounding boxes.

This process isolates the region of interest, ensuring that only facial region is considered for further processing. Reduces unnecessary computation and improves efficiency. This module is capable of detecting multiple faces in a single frame and processes each detected face individually. All of this happens in real time enabling identification as they appear in front of the camera.

Accurate face detection is essential to the performance of the system, as any error in this stage may affect the future processes.

4.13.3 Feature Extraction Module

The feature extraction module is responsible for extracting important facial features from the detected face and convert them into numerical format suitable for identification. These numerical formats are known as face encodings.

face_recognition library is used for feature extraction purpose. This library analyses the facial structure and converts them into face encodings.

The generated face encodings act as a digital signature for the individual as it is used for comparison during the recognition process. These encodings are compact and efficient for storage and processes unlike raw image.

This module plays a critical role in ensuring accurate identification. Equality of the extracted features directly affects the performance of the facial recognition system

4.13.4 Face Recognition Module

The face recognition module is responsible for identifying individuals by comparing the extracted face encodings with the stored face encodings in the database. This module checks whether the detected face belongs to authorized person or not

The face_recognition library is used to perform this process. This module compares the generated face encoding with stored face encoding using distance-based methods. It defines a threshold, ensuring only faces with similarities above that threshold will be granted access.

The system also retrieves the corresponding details of the authorized person from the database. If there is no match for that individual, the system labels that individual as unauthorized and alert the security. This process happens in real time, allowing the system to quickly identify individual as they appear in front of the camera.

The accuracy of recognition module is critical for secure entries in the defence areas as any inaccurate identification may lead to unauthorized entry.

4.13.5 Database Management Module

The database management module is responsible for storing and managing all the data required for this face recognition system. It stores the facial encodings and the details of authorized personnel.

In the proposed system, face encodings are stored in a pickle file which allows fast loading and efficient comparison. There are unique facial features of individuals that are used for identification.

The system uses SQLite database to store user information. When the face is recognized, the system also retrieves the corresponding details from the database and displays it on the interface. A log of all access attempts including entry and exit times; identity of individual and access status is maintained.

Efficient database management ensures quick data retrieval, smooth system performance, and easy updating of records making it essential part of the overall system.

4.13.6 Access Control Module

The access control module is responsible for making the final decision of access being granted or denied to an individual into the restricted area. This module uses the results obtained from the face recognition module to determine the access.

When a face is successfully recognized, the system retrieves the corresponding detail of that individual from the database.

If the individual is authorized, the system will grant access and display a confirmation method such as “Access Granted” for that individual. If the face is not recognized, the system denies the access and displays “Access Denied” message.

In case of unauthorized access, the security will be notified. Additionally, all excess attempts are recorded in a database log with details such as timestamp and access status for monitoring.

4.14 Frontend Implementation

The frontend of this facial recognition system is a combination of HTML, CSS and JavaScript, working together enabling smooth interaction between the user and the system. The frontend of the system provides an interactive and user-friendly interface for controlling the application.

HTML is used to create the basic structure of the web interface, including elements such as buttons, video display areas and result panels.

CSS is used to enhance the visual appearance of the interface but defining layout, colours and font. This ensures that the interface looks clear and easy to use.

JavaScript is used to add functionality to the system full It enables actions such as starting the camera, capturing frames and sending data to back end for processing.

Flask framework is used to connect the front end to the back end through HTTP request, allowing real-time communication between user interface and processing modules.

The frontend implementation enhances usability and ensures that the system is easy to operate in real-time environments.

4.15 Backend Implementation

The back end of the proposed system is responsible for processing data, performing face recognition and managing communication between components of the system. It is develop using Python and flask framework.

The backend handles request from the frontend interface. When a user initiates a scan through the web interface, a request is sent to the Flask server. Frames are captured from the camera and pass through image processing module.

OpenCV library is used to perform face detection on the captured frames. Once a face is detected, face_recognition library is used to extract facial features and generate face encodings. These captured encodings are compared to stored encodings loaded from the pickle file

If a match is found, the system will grant access to that individual and details related that individual will be retrieve from the database to the frontend. If no match is found, the system will identify that individual as unauthorized and inform the security.

The Flask framework ensures smooth communication between frontend and backend, allowing real time updates and quick response to the user actions.

4.16 Working of the System

The working of the proposed facial recognition system describes the complete process from capturing the image to granting or denying access. This operates in real time and quickly identifies individual as soon as they come in front of the camera.

The process begins with the camera capturing live video and extracting frames from that video. These frames are processed to detect the presence of faces using OpenCV library.

Once a face is detected, the system extracts the facial region and convert into face encoding using the face_recognition library. This encoding is compared to the stored encodings of authorized personnel loaded from the pickle file.

If a match is found, the system will grant access to that individual and the details corresponding to that individual will be retrieved from SQLite database. Otherwise, access will be denied and security will be informed.

All attempts whether successful or unsuccessful are recorded in the database. The system monitors the video stream and repeats this process each new frame, enabled in real-time operation.

This chef ensures efficient performance and accurate identification in the proposed system

Summary

In this chapter, the implementation of the proposed facial recognition system has been discussed. The technology is used for developing this system are also explained. The frontend and backend interaction has been highlighted. Overall, this chapter highlights the practical working and efficiency of the system.

Chapter – 5

Results and Discussion

This chapter presents the output of the proposed facial recognition system and evaluates its performance. This chapter demonstrates how the system works in real time by capturing image, detecting faces and recognizing individuals and granting or denying access on the verification results.

The results include screenshot of the web interface, face detection process, recognition system and system logs. These outputs help in understanding the effectiveness and reliability of the system in real world scenarios

5.1 System Output Screens

5.1.1 Web Interface

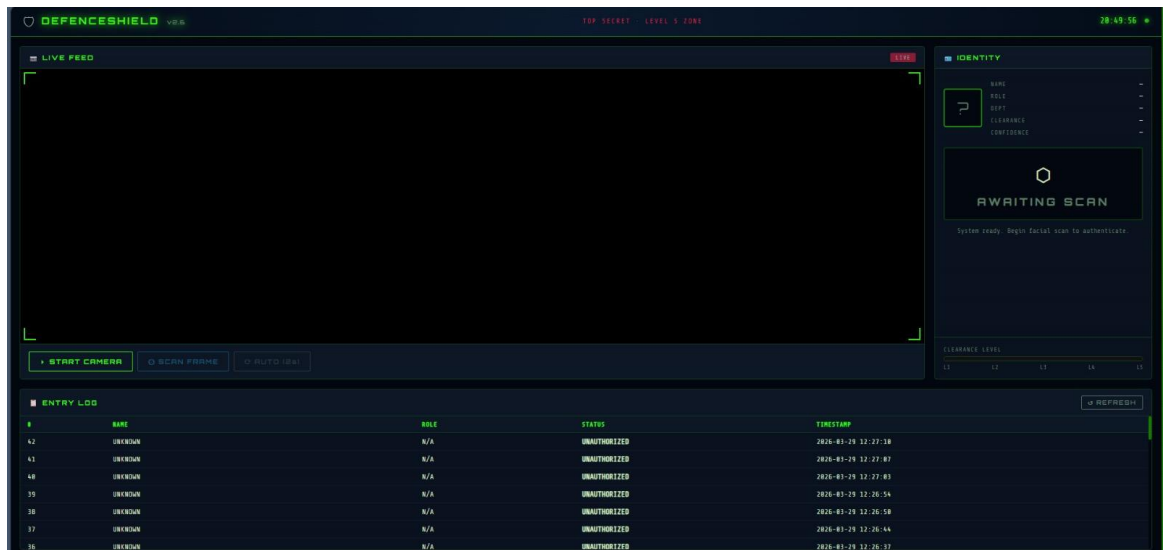


Figure 5.1: Web Interface of the System

This figure shows the web-based interface of the system developed using HTML, CSS, and JavaScript. It provides options to start the camera and perform face recognition.

5.1.2 Camera Activated

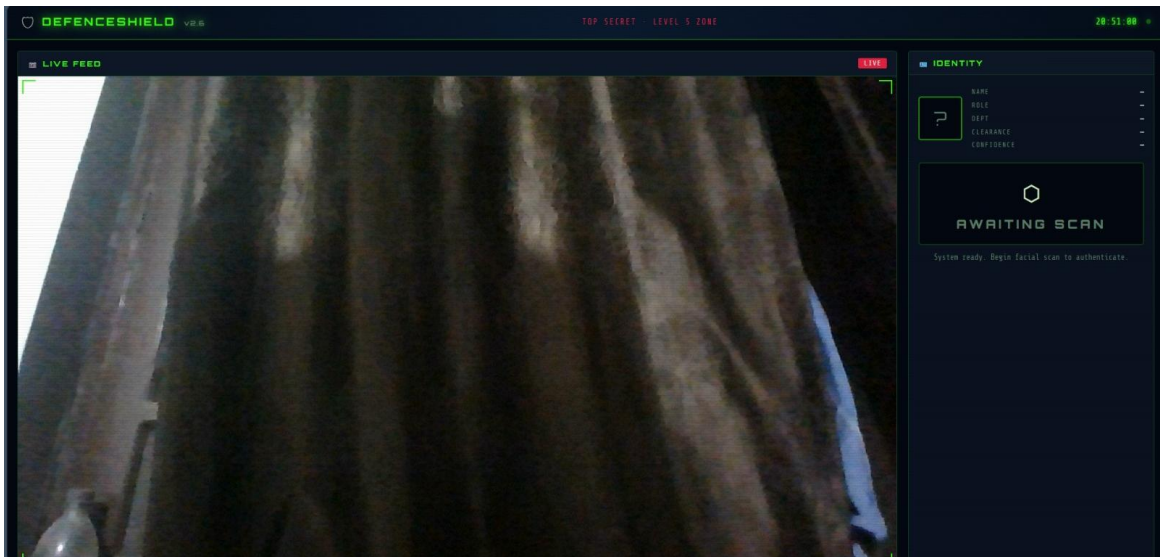


Figure 5.2: Camera Activated for Face Capture

This figure shows the camera interface when activated. The system captures real-time video frames for further processing.

5.1.3 Access Granted

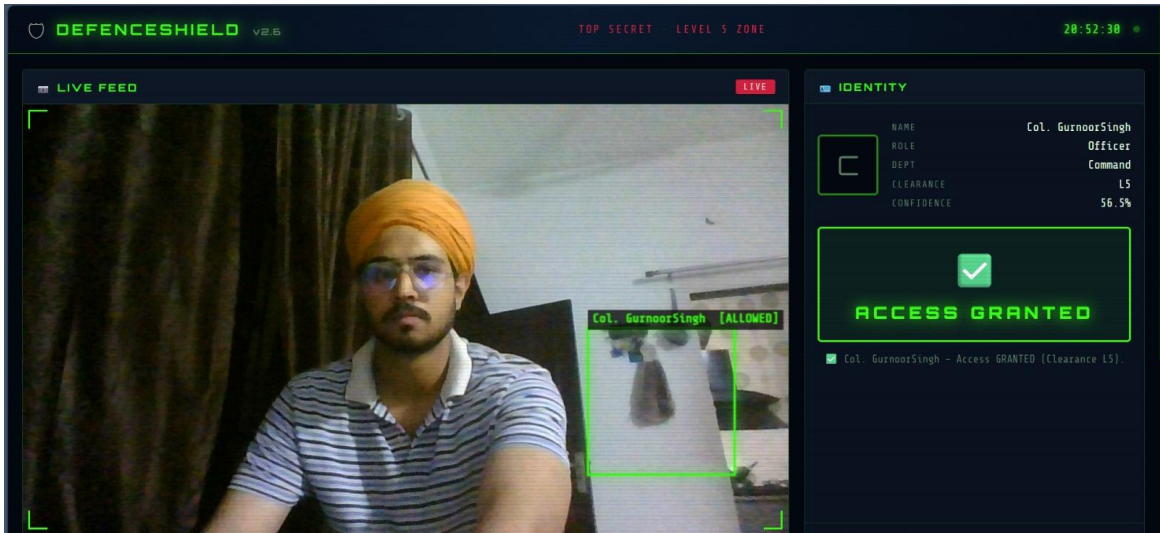


Figure 5.3: Authorized User Identified (Access Granted)

This figure shows the successful recognition of an authorized user. The system matches the face with stored data and grants access.

5.1.4 Access Denied

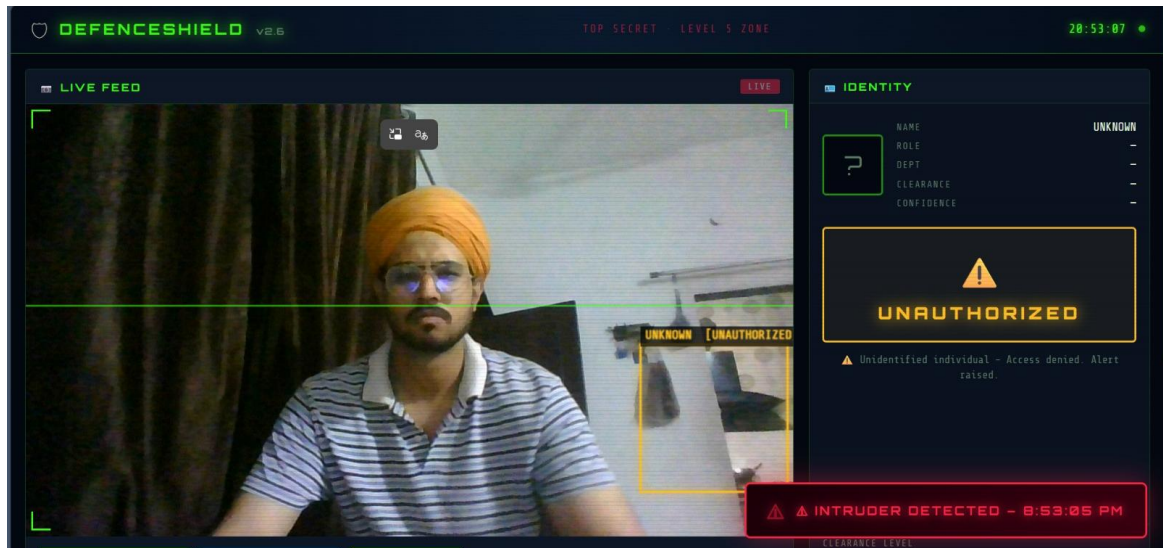


Figure 5.4: Unauthorized User Detected (Access Denied)

This figure shows the detection of an unauthorized user. Since no match is found in the database, access is denied.

5.1.5 Entry Logs

#	NAME	ROLE	STATUS	TIMESTAMP
49	Col. GurnoorSingh	Officer	ALLOWED	2026-04-23 15:23:21
48	Col. GurnoorSingh	Officer	ALLOWED	2026-04-23 15:23:18
47	UNKNOWN	N/A	UNAUTHORIZED	2026-04-23 15:23:13
46	UNKNOWN	N/A	UNAUTHORIZED	2026-04-23 15:23:09
45	UNKNOWN	N/A	UNAUTHORIZED	2026-04-23 15:23:05
44	UNKNOWN	N/A	UNAUTHORIZED	2026-04-23 15:23:02
43	Col. GurnoorSingh	Officer	ALLOWED	2026-04-23 15:22:26

Figure 5.5: Entry and Exit Logs Maintained by the System

This figure shows the log records maintained by the system, including details such as user identity, timestamp, and access status.

5.2 Result Analysis

The proposed facial recognition system was tested under various conditions to its performance and reliability. The system captures real time video, detect faces and perform recognition in real time.

The system accurately identifies authorize users by matching the captured phase encoding with the data in the database. When a valid match is found, access is granted instantly. In case there is no match, the identify the individual as unauthorized and deny access.

The system also maintains entry and exits log. Each attempt whether successful or unsuccessful is recorded in that log. This helps in the monitoring and analysing process. The response time of the system is fast, enabling real-time identification.

However, System Chill have some limitations We'll stop the accuracy of the system slightly decrease under poor lighting conditions or when the face was partially covered.

Overall, the indicate that the proposed system is efficient reliable and suitable for real time applications. It is secure and the process is automated which reduce human effort and improve overall security.

Summary

In this chapter, the results of the proposed facial recognition system have been presented. The system outputs, including web interface, face detection, recognition results and access control decisions were demonstrated using screenshots. The system has successfully identified authorized and unauthorized individuals. The system also maintains logs of entry and exit activities. Overall, the system proves to be efficient for secure automated access control.

Chapter – 6

Conclusion and Future Scope

6.1 Conclusion

The facial recognition system for secure entry and defence zone has been successfully designed and implemented. The system effectively uses computer vision and ML techniques to detect, recognize and verify individuals in real time.

The integration of technologies such as OpenCV, face_recognition and Flask have enabled the development of a complete system that includes both backend and frontend. The system includes interactive and user-friendly web-based interface through which the user can access it. The system is capable of capturing facial images, extracting features and comparing them to restore data in the database.

It provides a reliable and automated solution for the access control by reducing errors and overcoming other limitations of traditional methods. It ensures that only authorized individuals are granted entry, thereby increasing the security restricted areas.

6.2 Future Scope

- The proposed system can be further improved and enhanced by incorporating additional features and technologies.
- Some of the future enhancements are as follows:
- **Multi-Factor Authentication:** We can combine the facial recognition system with other authentication methods such as fingerprint or OTP for increased security.
- **Anti-Spoofing Techniques:** Implementing liveness detection to prevent spoofing attacks using photos or videos
- **Improved Accuracy:** Using advance deep learning methods to enhance recognition accuracy under challenging conditions.
- **Cloud Integration:** Storing data on cloud platforms for better scalability and remote access.
- **Large-Scale Deployment:** Expanding the system for use in large organizations with a large number of users.
- **IoT Integration:** The system can be integrated with IoT devices such as smart locks, automated gates and surveillance cameras.

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